

[illegible]

NOTE: STANDARD PPE OF SAFETY FOOTWEAR, SAFETY HELMET, COVERALLS, GLASSES AND GLOVES MUST BE WORN. ANY ADDITIONAL PPE REQUIRED SHALL BE CAPTURED IN THE MITIGATING CONTROLS SECTION OF THE TRA TEMPLATE BELOW.

No.	Task /Activity	Hazard and Consequence		Initial Risk			Mitigating Controls	Residual Risk		
	Sequential Steps	Hazard	Consequence	Probability	Consequence Severity	Risk Rating	Measures to Reduce Risk to ALARP	Probability	Consequence Severity	Risk Rating
1	Review TRA and TBT	Miscommunication	Personnel Injury / Environmental Pollution	1	2	M	Situational awareness and proper use of PPE. Hard hats, impact gloves, cotton gloves, latex gloves, steel toed boots, safety glasses, side shields and face shield	1	1	L
2	Prior to any task being undertaken ensure that all personnel are aware of the hazards involved.	Electrical shock, possible slips trips and falls. Pinch points poor body position. Hydraulic discharge and spills.	Personnel Injury / Environmental Pollution	2	3	M	All personnel must be aware of their surrounds and weather conditions. Use proper Manual Handling techniques, i.e., Always use proper body positioning, lift with legs, and obtain second personnel when loads are over 40lbs. Read all SDS on chemicals used, use PPE that is specified in SDS and be aware of where all spill kits are located.	2	2	M
3	Check weather condition prior ROV leaving the deck	Damage to equipment from weather patterns	Personnel Injury / Environmental Pollution / Damage to Equipment	2	3	M	Communication with bridge personnel prior to lifting system from deck.	1	1	L
Non-Personnel are not to be allowed around ROV during this operation										
4	Connect tag lines to ROV/ TMS for positive control.	Damage to equipment from weather patterns, slips trips and falls	Personnel Injury / Damage to Equipment	2	3	M	Constant communication and controlling ROV and TMS with tag lines.	1	1	L
5	Energize HPU on winch and ensure that cooling system on and is operational.	Damage to Winch HPU from overheating	Personnel Injury / Environmental Pollution / Damage to Equipment	2	3	M	Cooling system is on.	1	1	L
6	Ensure that grounding strap has been removed from TMS, and cargo straps have been removed.	Electrical shock, damage to equipment	Personnel Injury / Environmental Pollution / Damage to Equipment	2	2	M	Ensure that TMS is not powered up without grounding strap on.	1	1	L

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7	Winch operator will bring up TMS off stands and into docking head latches.	Damage to tether and equipment.	Personnel Injury / Environmental Pollution / Damage to Equipment	3	3	M	Winch operator will be in constant communication with observer	1	2	L
8	Winch operator will lace TMS into docking head latches and close latches.	Burst hydraulic hose, leak of hoses, hydraulic discharge.	Personnel Injury / Environmental Pollution / Damage to Equipment	3	3	M	Winch operator will be in constant communication with Pilot to unlatch TMS, and not 'Stretch' the tether. Pilot will only pay out only what is needed to complete task.	1	2	L
9	Winch operator will Erect/Luff LARS to bring TMS in proper position over ROV.	Equipment Damage, burst hydraulic hose on winch, leak of hoses.	Personnel Injury / Environmental Pollution / Damage to Equipment	3	3	M	Operator and observer must communicate to ensure that ROV is aligned properly and will not damage equipment. Have spill kit available.	1	2	L
10	Pilot will turn on TMS IPDS.	Electrical shock	Personnel Injury / Environmental Pollution / Damage to Equipment	2	2	M	Observer will ensure Pilot that on one is around TMS before activating system.	1	1	L
11	Pilot will enable TMS and begin to tether in extra tether as winch operator brings TMS over ROV.	Damage to tether and equipment, electrical shock. Burst hydraulic hose on winch leak of hoses and hydraulic oil discharge.	Personnel Injury / Environmental Pollution / Damage to Equipment	2	2	M	Pilot, observer, and winch operator will be in constant communication as tether is being pulled into the TMS as not to "stretch" the tether.	1	1	L
12	Winch operator will unlatch TMS from docking head and will begin to slowly lower TMS over ROV bullet	Burst Hydraulic hose, leak of hoses, hydraulic discharge. Pinch points and damage to equipment, and tether	Personnel Injury / Environmental Pollution / Damage to Equipment	3	3	M	Pilot, observer, and winch operator will be in constant communication as tether is being pulled into the TMS as not to "stretch" the tether.	1	2	L
13	Pilot will Latch TMS onto ROV bullet	Damage to equipment and tether	Personnel Injury / Environmental Pollution / Damage to Equipment	3	3	M	Pilot will communicate with winch operator when TMS has been latched into the ROV bullet.	1	2	L
14	Observer will unlatch cargo straps from ROV and remove ground strap.	Electrical shock	Personnel Injury / Environmental Pollution / Damage to Equipment	2	2	M	Ensure that ROV is not powered up without grounding strap on.	1	1	L
15	Winch operator will bring ROV off deck	Pinch points while removing stands from ROV.	Personnel Injury / Environmental Pollution / Damage to Equipment	2	2	M	Observer will remove the stands one at a time ensuring that they do not place themselves in a pinch point.	1	1	L

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16	Winch operator will slowly bring the ROV/TMS into the docking head and open the latches and latch ROV/TMS into docking head.	Burst Hydraulic hose, leak of hoses, hydraulic discharge.	Personnel Injury / Environmental Pollution / Damage to Equipment	3	3	M	Winch operator and observer will monitor LARS system for leaks. Spill kit available and ready for use.	1	2	L
Steps 17 and 18 are for Stands										
17	Position ROV/TMS in proper location for stands	Burst Hydraulic hose, leak of hoses, hydraulic discharge.	Personnel Injury / Environmental Pollution / Damage to Equipment	3	3	M	Winch operator and observer will monitor LARS system for leaks. Spill kits available and ready for use.	1	2	L
18	Land TMS/ROV on stands by positioning docking head over stands.	Swinging TMS/ROV impacting LARS or personnel, equipment damage.	Personnel Injury / Environmental Pollution / Damage to Equipment	3	3	M	Observer must be in constant communication with winch operator. Observer must be aware of all pinch points while adjusting stands.	1	2	L
Steps 19 and 20 are without Stands										
19	Unlatch ROV/TMS from docking head and slowly lower to LARS deck.	Burst Hydraulic hose, leak of hoses, hydraulic discharge. Pinch points and damage to equipment.	Personnel Injury / Environmental Pollution / Damage to Equipment	3	3	M	Winch operator and observer will monitor LARS system for leaks. Personnel will keep in constant communications as ROV/TMS is being lowered.	1	2	L
20	Land TMS/ROV on LARS deck.	Swinging TMS/ROV impacting LARS or personnel, equipment damage.	Personnel Injury / Environmental Pollution / Damage to Equipment	3	3	M	Observer must be in constant communication with winch operator. Observer must be aware of all pinch points.	1	2	L
21	Observer will attach grounding strap to ROV and secure ROV to LARS.	Electrical shock, Damage to equipment.	Personnel Injury / Damage to Equipment	2	2	M	Ensure TMS is strapped down to LARS system with large approved, cargo straps.	1	1	L
TRA Approval		Name	Position	Signature		Date				
By signing, the Approver confirms that the TRA has been reviewed and considers the task acceptable to proceed with residual risk meeting the acceptance criteria of DO Risk Matrix.										

*Reference DO-HSE-FRM-140 for DeepOcean Risk Assessment Matrix (RAM) and Guidewords